Laboratory Course n° 3
–
Lens & Lighting

Objective

Taking into account the scene you have to image and features to be detected, you have to choose:

- Lens
- Lighting configuration to implement.

Equipment

- PC Computer
- IDS Digital Camera, USB Cables
- Industrial Lenses
- Extension Rings
- Lighting Devices
- Industrial Parts

Software

IDS software (Uye Cockpit)

Documentation

- User Manual
- Lectures on Machine Vision

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Lens

In order to carry out this step, consult the following documentation:

- Camera Brochure
- Lectures on Machine Vision

You have to observe the fork in order to perform surface inspection with the following specifications:

- Working Distance : 500 mm



Choose the right lens, by using the estimation formula in the course.

Mount the selected lens on the camera and check that the computation is OK:

1. Connect Camera cables.
2. Start Ueye cockbit or demo software.
3. Select the right camera and start Grabbing images.

Ask professor to validate once everything is OK.

Compute the obtained "Spatial Resolution"

Select different settings and study performances w/r to full or partial frame acquisition => Select "Properties" and navigate in tabs.

Give your conclusions on the Camera Link Standard

Extension Rings

Mount different extension rings and observe the influence on the acquisition.

Give your conclusions on the use of extension rings

The camera sensor model is UI-125xSE-C

1. Technical Specifications

The exact model should be verified using the label on the camera. The reference may correspond to the IDS UI-1250SE-C-HQ variant.

Specification	Value
Manufacturer	IDS Imaging
Sensor type	CMOS color
Sensor model	e2v EV76C570
Image resolution	1600 × 1200 pixels
Total number of pixels	1.92 megapixels
Sensor dimensions	7.2 × 5.4 mm
Pixel size	4.5 µm
Interface	USB 2.0
Maximum frame rate	Approximately 17.6 frames per second

2. Focal Length Calculation

To select a suitable lens, we can use the following approximate formula:

$$\boxed{f = \frac{WD \times S}{FOV}}$$

Where:

- f = focal length of the lens (mm).
- WD = working distance (mm).
- S = sensor dimension (mm).
- FOV = field of view, or the size of the scene to be observed (mm).

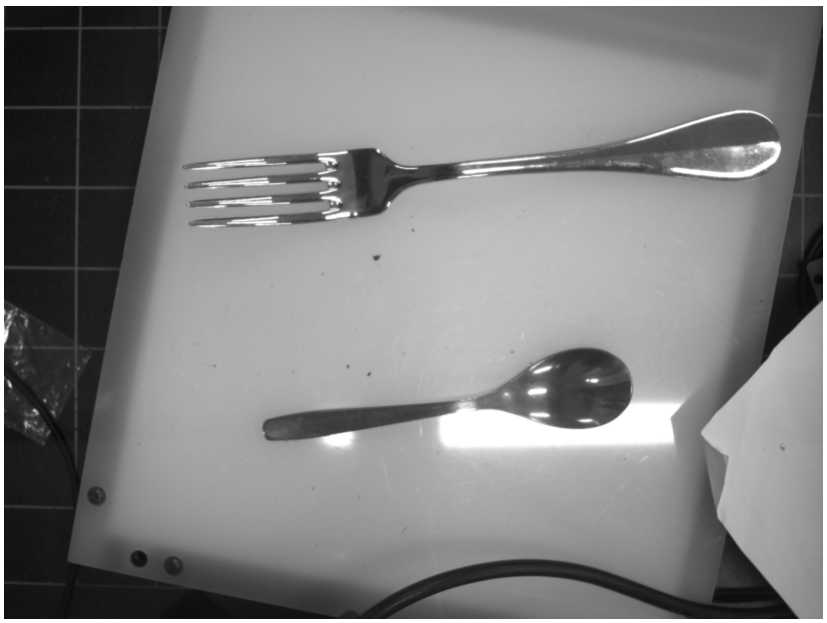
In this laboratory exercise, the working distance is **500 mm**.

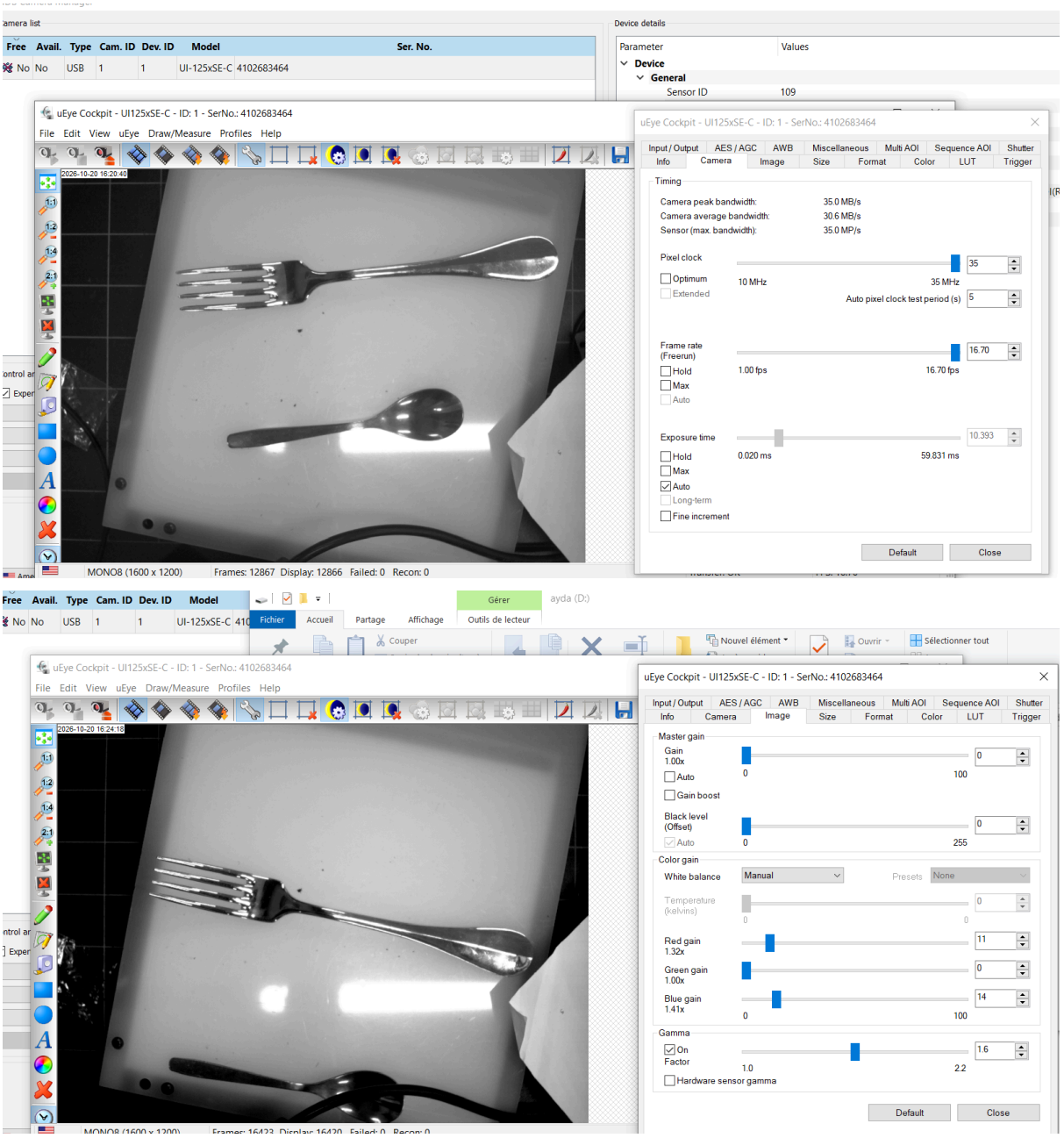
For example, if the required field of view is 200 mm horizontally:

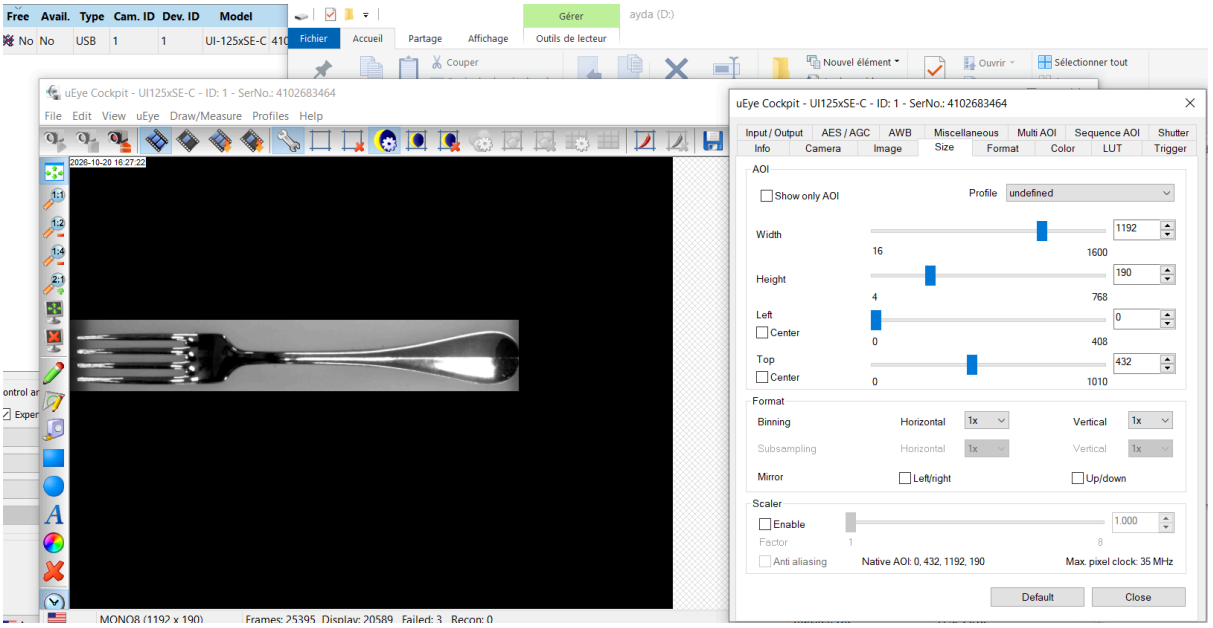
$$f = \frac{500 \times 7.2}{200}$$

$$\boxed{f = 18\ \text{mm}}$$

So we chose a 18 mm focal length and the resulted image is as follows
12 mm




amera list
Free Avail. Type Cam. ID Dev. ID Model Ser. No.
No No USB 1 1 UI-125xSE-C 4102683464
Device details
Parameter Values
Device
General
Sensor ID 109
uEye Cockpit - UI125xSE-C - ID: 1 - SerNo.: 4102683464
File Edit View uEye Draw/Measure Profiles Help
2026-10-20 16:20:40
MONO8 (1600 x 1200) Frames: 12867 Display: 12866 Failed: 0 Recon: 0
Input / Output AES / AGC AWB Miscellaneous Multi AOI Sequence AOI Shutter
Info Camera Image Size Format Color LUT Trigger
Timing
Camera peak bandwidth: 35.0 MB/s
Camera average bandwidth: 30.6 MB/s
Sensor (max. bandwidth): 35.0 MP/s
Pixel clock
Optimum
Extended
10 MHz
35 MHz
35
Auto pixel clock test period (s)
5
Frame rate (Freerun)
Hold
Max
Auto
1.00 fps
16.70 fps
16.70
Exposure time
Hold
Max
Auto
Long-term
Fine increment
0.020 ms
59.831 ms
10.393
Default
Close
Free Avail. Type Cam. ID Dev. ID Model
No No USB 1 1 UI-125xSE-C
Fichier Accueil Partage Affichage Gérer Outils de lecteur
Couper
Nouvel élément
Ouvrir
Sélectionner tout
uEye Cockpit - UI125xSE-C - ID: 1 - SerNo.: 4102683464
File Edit View uEye Draw/Measure Profiles Help
2026-10-20 16:24:18
Input / Output AES / AGC AWB Miscellaneous Multi AOI Sequence AOI Shutter
Info Camera Image Size Format Color LUT Trigger
Master gain
Gain
1.00x
Auto
Gain boost
0
100
Black level (Offset)
Auto
0
255
Color gain
White balance
Manual
Presets
None
Temperature (kelvins)
Red gain
1.32x
Green gain
1.00x
Blue gain
1.41x
11
14
Gamma
On
Factor
1.0
2.2
1.6
Hardware sensor gamma
Default
Close



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Where:

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In this laboratory exercise, the working distance is **500 mm**.

For example, if the required field of view is 200 mm horizontally:

$$f = \frac{500 \times 7.2}{200}$$

$$f = 18 \text{ mm}$$

Pixel size 4.5 micro meter and the width of the fork is around 1192 pixel:

The field of view visible to the camera is 29 cm and the image size is 1600 * 1200,
Which to calculate the Length of the real = L of pixels * Rs

Rs= FOV/N
Where N is the width of the image

We calculate the length of the fork to be 21 cm which is close to the real value.

Method 1: Using the spatial resolution (recommended for your TP)

The spatial resolution tells you how much real-world distance corresponds to one pixel.

Step 1: Calculate the spatial resolution

$$\boxed{R_s = \frac{FOV}{N}}$$

Where:

- R_s = spatial resolution (mm/pixel).
- FOV = field of view, the real-world width visible in the image (mm).
- N = number of pixels across that width.

Step 2: Measure the object in pixels

Suppose the object's length in your image is:

$$L_{\text{pixels}} = 1192 \text{ pixels}$$

Step 3: Convert to real-world length

Use:

$$\boxed{L_{\text{real}} = L_{\text{pixels}} \times R_s}$$

Example: Suppose your field of view is 200 mm across 1600 pixels.

First, calculate the spatial resolution:

$$R_s = \frac{200}{1600} = 0.125 \text{ mm/pixel}$$

Then calculate the object's real-world length:

$$L_{\text{real}} = 1192 \times 0.125$$

Ask ChatGPT Think